P07

Modal Analysis and Orthogonality in Multi-Degree Systems

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Published Modal Testing: Theory and Practice (1984)

Paper Link https://doi.org/10.1115/1.3269255

Domain DOM_ENG

Abstract

Multi-degree-of-freedom structures possess coupled mass and stiffness matrices. Generalized eigenvalue problems yield orthogonal mode shapes that diagonalize system matrices into decoupled modal oscillators.

Abstract quoted from D. J. Ewins et al., Modal Testing: Theory and Practice (1984).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?